

Estimating Number of Computations in Blockchain Mining

In Bitcoin mining, we can estimate the number of hash computations required to find a valid block using the following formula:

$$P \times 10 \text{ min} \times 60 \text{ sec} = 2^n$$

$$n = \log(600P) / \log(2)$$

Explanation of Terms:

P = Hash rate (hashes per second)
600 = Total seconds in 10 minutes (average block time)
 2^n = Expected number of hash attempts required
n = Difficulty in terms of bits

Example:

Let $P = 10^{20}$ hashes/second

Total computations:
 $P \times 600 = 10^{20} \times 600 = 6 \times 10^{22}$

Convert to power of 2:
 $n = \log_2(6 \times 10^{22}) \approx 75$

Result:
Total computations $\approx 2^{75}$ hashes

Summary:

1. Multiply hash rate by 600 seconds to get total computations.
2. Convert result into power of 2 using logarithm.
3. The value n represents mining difficulty in bits.

Higher n $\rightarrow$ More computations $\rightarrow$ Higher difficulty